



Tribhuvan University
Institute of Engineering
Pulchowk Campus

Indian Institute of Technology
Bombay
Mumbai, India

Internship Report

On

Propellant Combustion Studies

SUBMITTED IN PARTIAL FULFILLMENT OF THE REQUIREMENT FOR THE BACHELOR'S DEGREE IN AEROSPACE ENGINEERING

Submitted By:
Raju Kumar Sah
Roll No: 078BAS034
Department of Mechanical and Aerospace Engineering
Pulchowk Campus, Institute of Engineering
Tribhuvan University

Submitted To:
Prof. Nagendra Kumar
Department of Aerospace Engineering
Indian Institute of Technology Bombay
Mumbai, India

Executive Summary

This internship was done at the Propellant Research Laboratory Indian Institute of Technology Bombay and was focused on experimental combustion studies related to solid, slurry, gel, and hybrid rocket propulsion systems. The work includes experimentation and observation of multiple propellant combination and their characterization.

The experimental activities included laser ignition studies of AP-HTPB composite solid propellants, Crawford Burner (CB) tests of ECSP and slurry-based propellants, and the synthesis and combustion testing of organic and inorganic gel propellants. Also, the assembly, and firing of an HTPB-N_2O hybrid rocket motor were also done and observed, providing information to hybrid propulsion system integration and testing procedures. With the involvement in the different experimentation, familiarity with the instrumentation and the software is achieved.

Objective on understanding combustion characteristics, ignition characteristics, flame, and safety considerations associated with handling propellant materials is done. Various combustion study techniques were used to analyze ignition delay, flame structure, and burning behavior under controlled laboratory conditions. The internship provided practical experience in propellant preparation, experimental setup, data acquisition, and laboratory safety protocols.

During the internship period, several outcomes were achieved including successful characterization of multiple propellant formulations, improved understanding of propellant behavior under different conditions, practical experience with advanced instrumentation and measurement techniques, and contribution to ongoing research activities within the department.

ACKNOWLEDGMENT

I would like to express my sincere gratitude to the **Propellant Research Laboratory (PRL)**, Department of Aerospace Engineering, IIT Bombay, and **Prof. Nagendra Kumar** for providing me with the opportunity to undertake this internship and gain valuable hands-on experience in propellant research and testing.

I extend my deepest appreciation to the research scholars **Mr. Deepak Govindaraj, Mr. Sachin Sonage, Mr. Yashwant KM, and Mrs. Harpreet Kaur**, whose guidance and expertise were extremely helpful in understanding the intricacies of solid propellant combustion and performance evaluation. Their insightful discussions and constant support greatly enhanced my learning experience.

I would also like to extend my heartfelt appreciation to **Asst. Prof. Dr. Sudip Bhattrai** and **Asst. Prof. Kamal Darlami**, whose efforts in facilitating this internship made this experience possible. Their coordination and support in arranging this opportunity played a crucial role in my learning journey.

Lastly, I would like to thank my peers and fellow interns for their cooperation and teamwork throughout this internship. This experience has significantly contributed to my technical knowledge and practical skills in propulsion research, and I deeply appreciate everyone who played a role in making this internship a rewarding and valuable experience.

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1. Introduction

Propellants are high-energy materials that undergo rapid chemical reactions to release energy, thereby generating the thrust required for rocket and aircraft propulsion. They form the backbone of any propulsion system, and their performance directly influences mission efficiency, reliability, and overall success. Consequently, a detailed study of propellant properties and behavior is essential for the design and development of safe and effective propulsion systems. Based on their characteristics, propellants are broadly classified into solid propellants, liquid propellants, and hybrid propellants. Each category offers distinct advantages and is selected according to specific mission requirements. To ensure optimal performance, propellant characterization is carried out to determine the most suitable type and composition for a given application. This characterization involves systematic testing and measurement of critical properties such as burn rate, mechanical strength, thermal stability, and energy content. The Propellant Research Laboratory at IIT Bombay is equipped with advanced facilities for the formulation, testing, and evaluation of diverse propellant systems. During my internship, I gained hands-on experience in this laboratory, where I learned various propellant formulation techniques and experimental methods for property characterization. This exposure provided me with practical insights into the scientific and engineering principles underlying modern propulsion systems.

1.1 Objectives

During the course of the internship period the main objectives were:

- To Gain hands-on experience with laboratory systems, advanced test facilities, and professional work culture.
- To develop understanding of various instrumentation systems through practical exposure and laboratory engagement.
- To acquire knowledge of various propellant types and their applications in propulsion systems.
- To participate in experimental testing and data collection to support propellant characterization and performance evaluation.

- To understand safety protocols and procedures while handling chemicals.
- To understand the propellant chemistry and combustion.

1.2 Work Done

During the internship duration from September to October of 2025, I worked under the supervision of Prof. Nagendra Kumar at the Propellant Research laboratory, IIT Bombay. The scope of my work included:

- Work in preparation of different propellants types such as Hybrid, solid, slurry and gel.
- Literature review on different propellant systems and their characterization method.
- Participation in experimental testing for reduction of ignition delay in laser ignition.
- Work in formulation, testing and postprocessing of gel propellants.
- Crawford burn test of Slurry and ECSP propellants.
- Observation and work involvement test procedure in Hybrid motor testing.

The internship provided valuable practical exposure to propellant research and helped me connect the theoretical knowledge from my coursework to real world applications.

2. Laser Ignition of AP-HTPB Propellant

2.1 Background

The concept of using high-power lasers to ignite propellants and explosives remotely, without physical contact, has been explored as an alternative to conventional ignition systems. Experimental study have shown that high energy lasers operating in both visible wavelengths and infrared wavelengths (such as Nd:YAG lasers) are capable of initiating combustion[1]. Laser heating produces localized high-temperature regions, leading to rapid combustion and deflagration. Numerous studies have examined the interaction of laser energy with propellants and pyrotechnic compositions, including ignition thresholds and delay times for different materials.

Laser ignition of solid propellants offers improved control over the ignition process and enhanced operational safety; however, achieving sustained ignition in practice is often difficult because many propellant formulations exhibit low optical absorptivity at commonly available laser wavelengths. Conventional opacifiers can increase absorption, but they are generally inert and do not contribute to the chemical reactivity of the propellant. It is studied that this limitation can be adressed by optically sensitizing ammonium perchlorate (AP)/hydroxyl-terminated polybutadiene (HTPB) composite propellants through the incorporation of reactive nano-aluminum (nAl)/polyvinylidene fluoride (PVDF) composite particles[2].The presence of PVDF promotes additional gas generation, while rapid combustion of aluminum particles near the surface increases burning rate and pressure sensitivity.

2.2 Methodology

Study and experimentation was done on laser-assisted ignition of Ammonium Perchlorate (AP) + Hydroxyl-Terminated Polybutadiene (HTPB) composite propellant.The composition consisted of an additives specifically nano aluminum (nAl) to study the ignition delay behavior. Tasks included sample preparation, positioning under the laser test bench, and recording ignition delay behavior. The test was done under different ambient pressure condition and various laser power. Fig 2.1 shows the experimental setup for

the laser ignition propellant chamber which is equipped with multiple sensors, including a piezoelectric sensor, photodiode, and pressure transducer for detection of ignition and pressure rise.

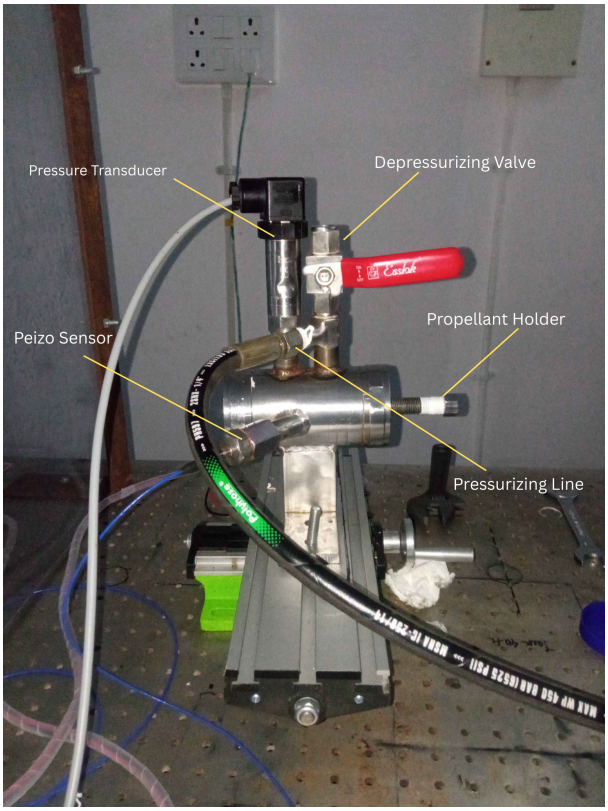


Figure 2.1: Laser ignition setup

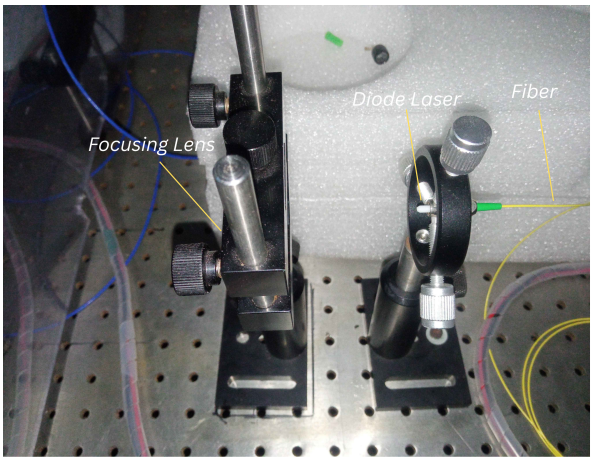


Figure 2.2: Laser diode

The test was performed under various ambient pressure condition which is pressurized with the nitrogen feed line, to study the effect of propellant laser ignition under high pressure condition. The ignition was done varying the multiple parameter like laser power, laser on delay, ignition sequence procedure etc.

2.3 Discussion

The laser ignition measurement results presented in Table 2.1 show the relations between applied laser power, on duration, and the measured laser-on time and piezoelectric sensor delay. At low laser power levels (10-20%), ignition was not consistent. At 40% power the ignition became quite consistent with laser on times and piezo delays. At higher powers (60-100%), ignition was faster and more consistent, showing reduced delay times.

Table 2.1: Laser ignition measurements

S. N	Power (%)	Time (ms)	Laser ON (ms)	Piezo Delay (ms)
1	10	80	–	–
2	20	80	–	46
3	20	50	63	55
4	20	40	54	–
5	20	40	55	76
6	40	30	40	62
7	40	30	43	37
8	40	20	34	32
9	40	10	17	50.17
10	60	8	15	17.5
11	60	8	15	20
12	100	5	12	8
13	100	5	–	–

Figure 2.3 shows the time signals recorded during the laser ignition experiment. The working channel A1 represents the laser driver voltage, while channel A0 represents the photodiode output measuring optical emission from the propellant combustion.

The laser driver voltage profile(A1) exhibits a rapid rise showing the working of the laser pulse with a region region that represents the active firing interval of the laser. After the interval, the driver signal decreases back toward its normal level, indicating laser shutoff.

The photodiode signal (A0) remains near its normal line until the laser activation and then gives a sharp response immediately after the driver volatge begins, confirming optical sense and ignition light.

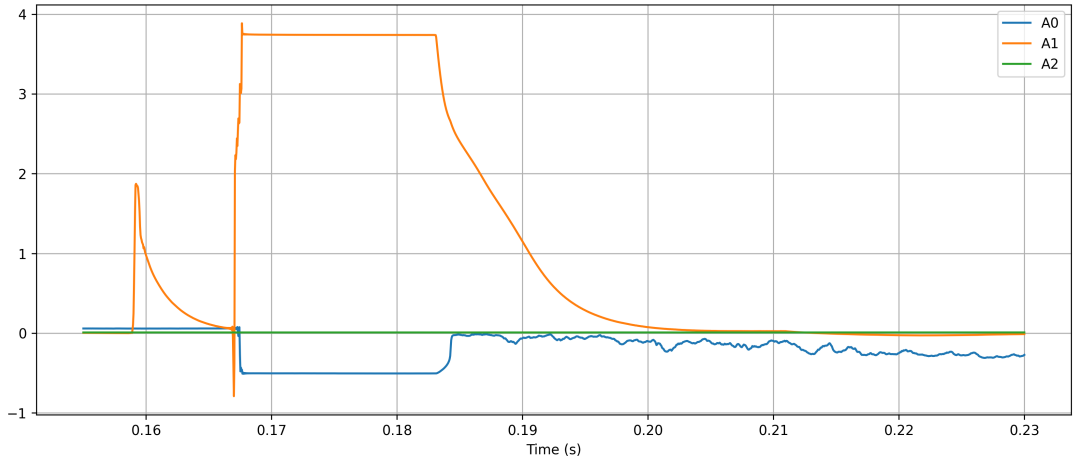
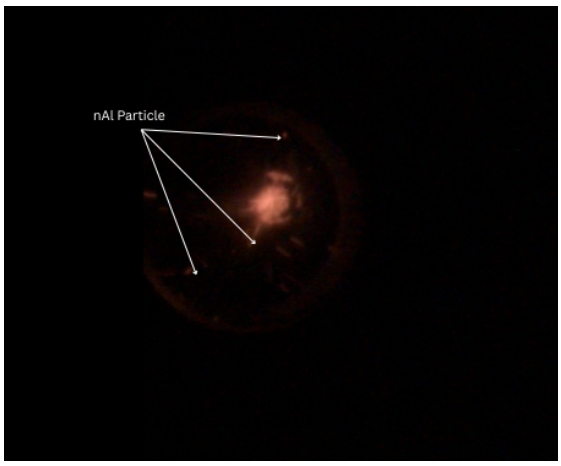


Figure 2.3: Laser ignition data analysis at 5 bar, nAl, 7.5 A



(a) Ignition initiation



(b) Ignition with Combusted nAl Particle



(c) Ignition with pyrolysis



(d) Ignition shutoff

Figure 2.4: High speed footage of laser ignition

3. Crawford Burner Test of Electrically Controlled Solid Propellant

3.1 Background

Electrically Controlled Solid Propellants (ECSP) are an innovative class of solid propellants that enable ignition, shutdown, and thrust modulation through electrical signals, offering precise control over rocket motor operation and enhanced flexibility in propulsion systems. Unlike conventional solid propellants, ECSPs allow regulation of start-up and burning rate by adjusting the supply voltage, making them highly responsive to mission requirements. Designing efficient ECSP formulations is complex, requiring optimization of burning rate, energy density, mechanical strength, and processability while ensuring operational reliability. As part of the work, involvement was made in Crawford burn testing of ECSP propellants under controlled voltage conditions to study ignition and combustion behavior. Additionally, fundamental properties such as density, electrical conductivity, combustion performance, mechanical strength, and processability were investigated, contributing to a deeper understanding of ECSPs and their potential to revolutionize solid propulsion systems through real-time electrical control.

3.2 Methodology

A lithium perchlorate–based Electrically Controlled Solid Propellant (ECSP) formulation was evaluated for burn rate using a Crawford strand burner setup to characterize its combustion behavior under varying ambient pressure, supply voltage, and temperature conditions. Small propellant strands were mounted in the sample holder with their ends connected to the cathode and anode, enabling electrical ignition and controlled burning. To ensure one-dimensional regression, the side surfaces of the strands were coated with silica gel as an inhibitor, restricting combustion to the intended burning face. The strand burn tests provided reliable burn rate data and verified stable combustion response, and the results were used for burn rate characterization and performance evaluation of the ECSP formulation for advanced propulsion applications.

4. Crawford Burner Test of Slurry Propellant

4.1 Background

As defined composite solid propellants consists of 70–80% of oxidizers such as ammonium perchlorate (AP), 20-30 of powdery metallic fuel such as aluminum, and polymeric binders such as hydroxyl-terminated polybutadiene (HTPB) which play an important role in the military and aerospace fields as a working substance for rocket or missile engines. Typically, composite solid propellants are prepared and manufactured using mixing machines. In the manufacturing of solid propellants, the mixing process is the most important and dangerous process. Mixing could result in combustion, explosions, and other safety concerns.

4.2 Methodology

HTPB-based slurry propellant was prepared using a non-curing binder without isocyanate curing agents. The formulation consisted of Ammonium Perchlorate (AP) as oxidizer (79 wt%), Activated Charcoal as fuel additive (1 wt%), HTPB binder (17 wt%), and DOA plasticizer (3 wt%). Before processing, the working environment was maintained and relative humidity was controlled near 45 to limit oxidizer moisture absorption. Standard protective equipment was used, and all metallic equipment was grounded to minimize electrostatic discharge risk.

All equipment was cleaned before use. The mixer bowl and tools were cleaned with ethyl acetate to remove residues. The solid components were first prepared by weighing AP and Activated Charcoal separately and manually blending them to obtain a uniform dry mixture. The multiple AP particlsizes were proportioned ae s required before blending. Parallely, the liquid components was prepared by weighing HTPB and DOA into a beaker and stirring until a uniform liquid premix was obtained.

The dry mixture was then gradually introduced into the liquid premix with continuous manual stirring to ensure proper wetting of the solids and to avoid agglomeration. This preliminary slurry was transferred into a sigma mixer for mechanical homogenization.

The mixture was processed for approximately 45 minutes, with intermittent scraping of the walls and blades to provide uniform consistency. During processing the hot water bath is provide in a sigma mixer.

Following mixing, the slurry was transferred into the beaker Since no curing agent was used, the material remained in an uncured, semi-solid slurry state.

After completion, all equipment was cleaned promptly with suitable solvents, and the prepared slurry was stored in a sealed, moisture-resistant container and finally in the heated oven.

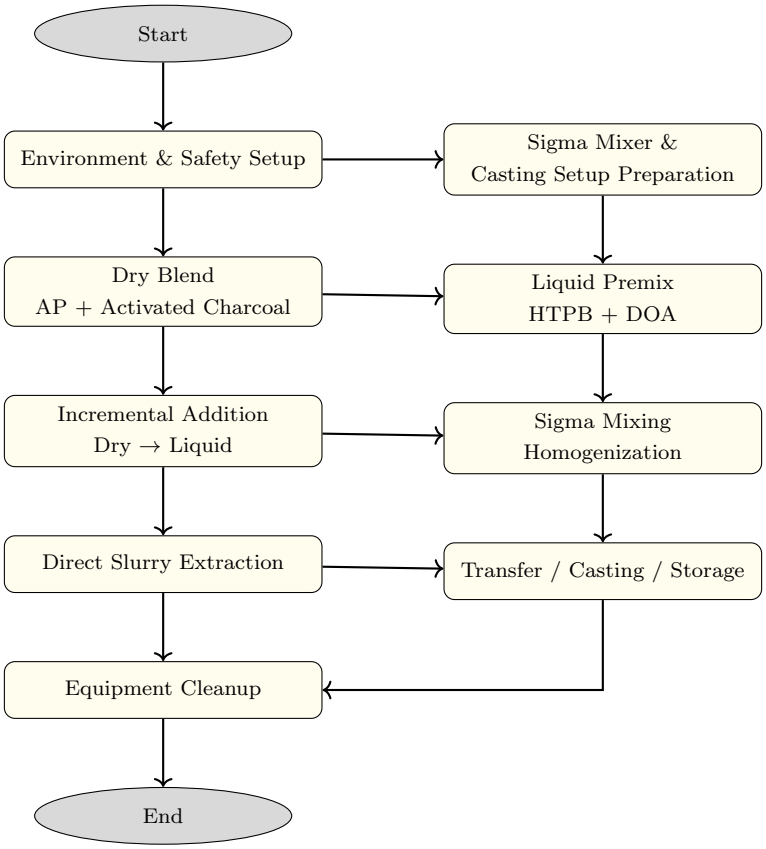


Figure 4.1: Flowchart of slurry propellant preparation process

4.3 Discussion

The prepared slurry propellant was tested in the CB setup. During testing, burn rate parameters were observed under varying ambient pressure and temperature condition. For test sample preparation the prepared slurry propellant was cast to small strand tube with a required length with a consistent diameter. It is desired to have uniform surface regression of the strand during burning of the propellant hence on inhibition surface silica grease was used.

The burn rate characteristics based on an ambient pressure was evaluated using Saint Robert's law, expressed as $r = aP^n$, where r is the burn rate, P is the pressure, and a and

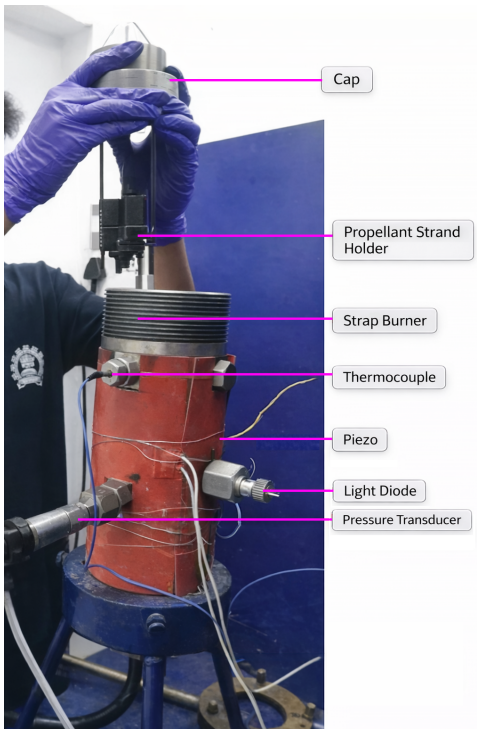


Figure 4.2: Crawford Setup

n are empirical constants. Experimental burn rate values were calculated from measured strand height and burn time for each test sample. A log–log transformation was applied to convert the relation into linear form for regression analysis to determine the pressure exponent and coefficient.

A linear fit of $\ln(r)$ versus $\ln(P)$ was performed using least squares regression. The slope of the fitted line gives the pressure exponent n, while the intercept gives $\ln(a)$. Multiple Crawford test was done with for different propellant combination. For the slurry propellant all the composition includes the 80% solid loading while the 20% of the mixer consists of binder. Table 4.3- 4.5 shows the burn rate parameter obtained under the pressure range of (20-70)bar for slurry propellant with different additives. The test was performed under the ambient room temperature.

Table 4.1: Burn rate Parameters $(20 - 70$ bar) for no additives

Parameter	**Value**	**Units**
Burn Rate Coefficient (a)	1.4159	$\text{mm}/(\text{s} \cdot \text{bar}^n)$
Pressure Exponent (n)	0.4433	-
Correlation Coefficient (R^2)	0.9784	-
Pressure Range	20-70	bar

On analyzing the data obtained from the test data, it is observed that all different propellant composition exhibits the same pressure sensitivity with the same approx pressure exponent value of $n = 0.4$ which is the typical for the AP-HTPB composite propellant.

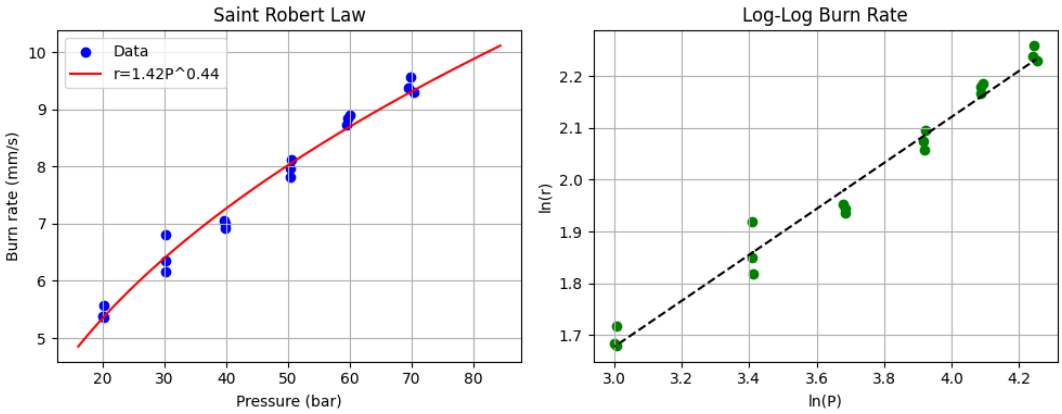


Figure 4.3: Burn rate relation for slurry propellant with no additives

Table 4.2: Burn rate Parameters $(20 - 70$ bar) for iron oxide additives

Parameter	**Value**	**Units**
Burn Rate Coefficient (a)	1.7524	$\text{mm}/(\text{s} \cdot \text{bar}^n)$
Pressure Exponent (n)	0.4098	-
Correlation Coefficient (R^2)	0.9099	-
Pressure Range	20-70	bar

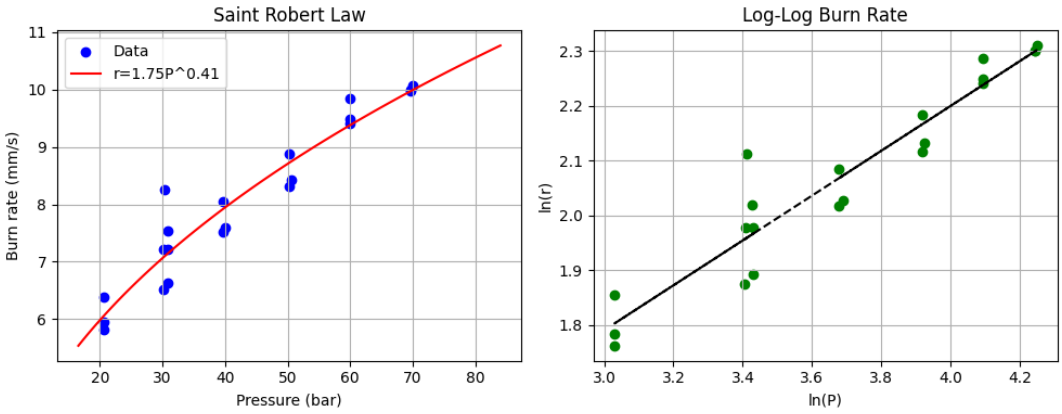


Figure 4.4: Burn rate relation for slurry propellant with Iron Oxide

Table 4.3: Burn rate Parameters $(20 - 70$ bar) for activated charcoal additives

Parameter	**Value**	**Units**
Burn Rate Coefficient (a)	1.5659	$\text{mm}/(\text{s} \cdot \text{bar}^n)$
Pressure Exponent (n)	0.4861	-
Correlation Coefficient (R^2)	0.9397	-
Pressure Range	20-70	bar

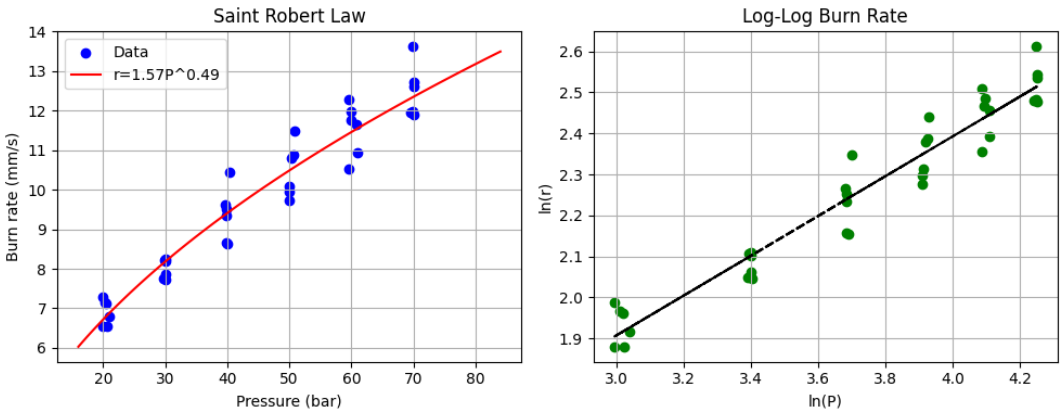


Figure 4.5: Burn rate relation for slurry propellant with Activated Charcoal

5. Gel Propellant Synthesis and Droplet Combustion Testing

5.1 Background

Gel propellants combine the advantages of liquid and solid fuels by offering reduced leakage risks, controlled flow and atomization, and improved combustion. Depending on the gellant and fuel, gel propellants are classified into inorganic and organic systems. Inorganic gel propellants typically inlcudes solid particulate gellants such as silica (SiO_2) and energetic additives like aluminum (Al) added to fuel like aviation turbine fuel to improve the rheological control and combustion performance. Similarly, organic gel propellants utilize organic fuels such as aviation turbine fuel (ATF), gelled using polymeric thickeners like Thixatrol, resulting in pseudoplastic behavior[3]. Study of the combustion behavior of these gel systems, particularly under varying droplet sizes and chamber pressures, is essential for their application in advanced propulsion systems.

5.2 Methodology

The organic gel propellant used in this study was a kerosene based aviation turbine fuel (ATF) containing xylene as a solvent and Thixotrol, and Silica as a gellant based on its type weather its organic or inorganic. In addition to organic gels, inorganic gel propellants were also formulated and evaluated. These propellants consisted of a liquid fuel thickened using silica (SiO2), with aluminum particles added in selected formulations to increase energy density. The compositions of the inorganic and organic gel propellants investigated in this work are listed in Table 5.1.

The both inorganic and organic gel propellant was prepared by carefully weighing it and mixing using using mechanical mixing process, with the mixing speed and duration selected based on the desired silica and aluminum mass content. Firstly, the liquid base fuel was poured into a clean mixing vessel and stirred at a low rotational speed to ensure uniform temperature and homogeneity. Organic and inorganic gellant was then added gradually while maintaining a stirring speed, allowing proper dispersion and preventing agglomeration.

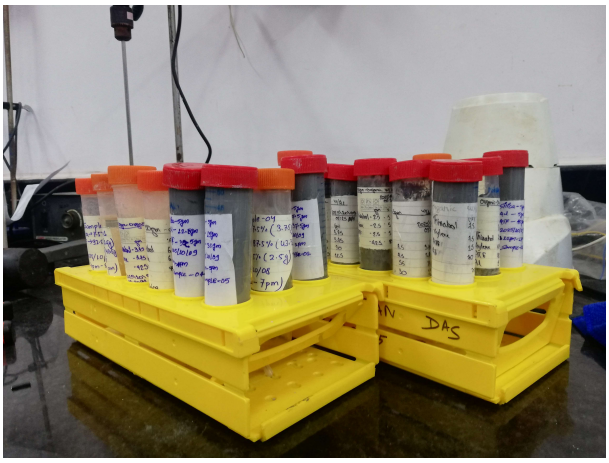


Figure 5.1: Prepared Gelled Propellant

Table 5.1: Inorganic Gel Propellant Compositions

S.N.	**SiO_2**		**ATF**		**Al**	
	%	**g**	**%**	**g**	**%**	**g**
1	5%	2.5g	95%	47.5g	0%	–
2	5%	2.5g	90%	45g	5%	2.5g
3	7.5%	3.75g	92.5%	46.25g	0%	–
4	7.5%	3.75g	87.5%	43.75g	5%	2.5g
5	10%	5g	80%	40g	10%	5g
6	10%	5g	75%	37.5g	15%	7.5g
7	10%	5g	70%	35g	20%	10g
8	10%	5g	65%	32.5g	25%	12.5g
9	10%	5g	60%	30g	30%	15g

The mixture was continuously stirred for the required total mixing time until a stable and homogeneous gel consistency was obtained. The prepared gel propellant was allowed to rest for a desired time before further testing or characterization.

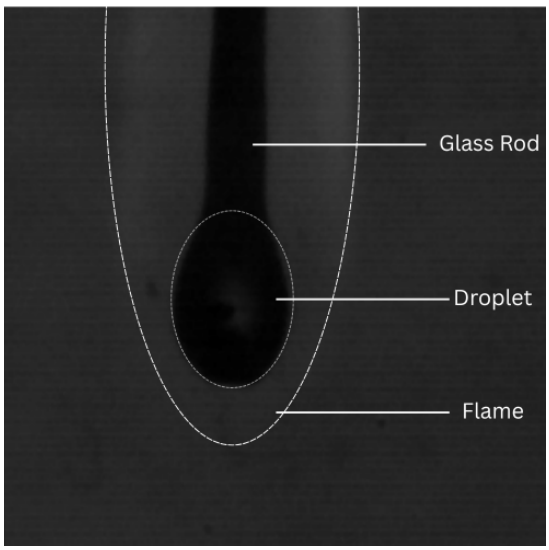
Table 5.2: Organic Gel Propellant Compositions

S.N.	**Thixotrol ST**		**Xylene**		**Al**		**ATF**	
	%	**g**	**%**	**g**	**%**	**g**	**%**	**g**
1	5%	2.5g	5%	2.5g	0%	–	90%	45g
2	6.5%	3.25g	6.5%	3.25g	0%	–	87%	43.5g
3	7.5%	3.75g	7.5%	3.75g	0%	–	85%	42.5g
4	7.5%	3.75g	7.5%	3.75g	15%	7.5g	70%	35g

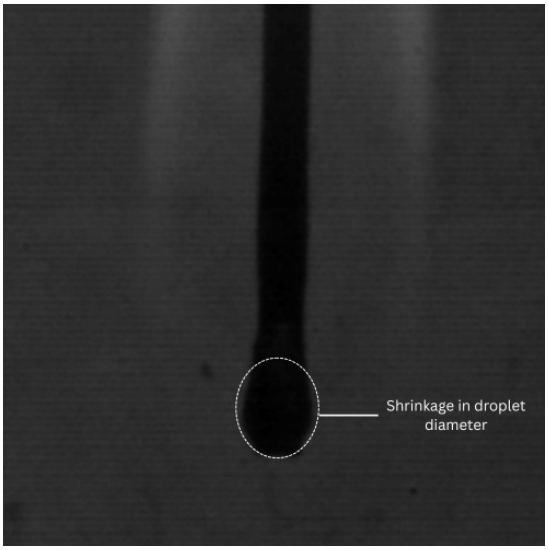
5.3 Discussion

The combustion of the gel propellants was done by the suspended burning in a glass under high speed camera, as shown in Figure 5.2. Figure 5.2(a) shows the combustion

phase, where flame is established around the droplet. As combustion progresses the fuel is consumed, leading to the reduction in droplet diameter.



(a) Formation of flame envelope



(b) Droplet surface regression

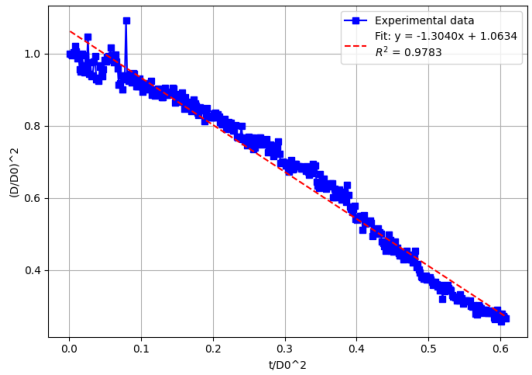
Figure 5.2: Stage of gel propellant combustion. (a) A flame surrounding the droplet during the combustion. (b) Significant reduction in droplet diameter observed during the final burn stage

The results shown in Fig. 5.3 shows a profile of inorganic gel propellant behavior on both silica (SiO_2) gellant loading and aluminum metal content. Increasing the silica fraction from 5% to 7.5% make the internal gel structure strong leading to higher viscosity. The lower silica formulations shows a more fluid like structure, providing faster reaction. The addition of 10% aluminum provides the energetic contribution of the gel. Compositions containing aluminum generally show higher energy release compared to non-aluminized gels. Overall silica primarily determines the rheological structure and combustion characteristics, while aluminum controls energy release.

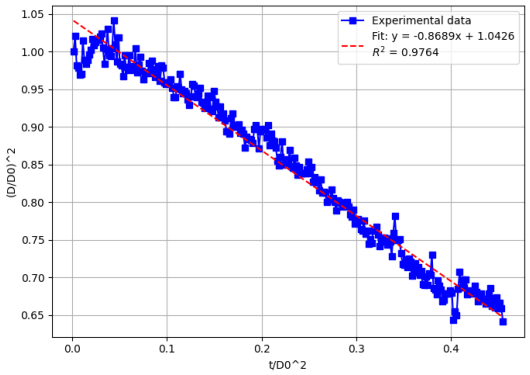
Figure 5.4 shows the combustion of gel propellant droplets with a fixed silica content of 10% SiO_2 and varying aluminum fractions (10% to 30%), analyzed using the D^2-law. Since the silica content is constant, the base gel is maintained, allowing for a comparison of the influence of metal fuel concentration on the droplet regression. The experimental data indicates that increasing the aluminum content significantly effects the regression rate. At lower aluminum content (10% and 15% Al), the droplets shows moderate regression rates. These lower content cases (Figs. 5.4a and 5.4b) also display significant chnage in the diameter, showing the unsteady burning behaviors.

However, as the aluminum content increases to 25% (Fig. 5.4c), a increase in the burning rate is observed. The regression slope also increase, and the total burning time decreases. The maximum R^2 value of 0.8998 is reached at at 25% Al composition. This shows that at this composition, the combustion process becomes more steady and con-

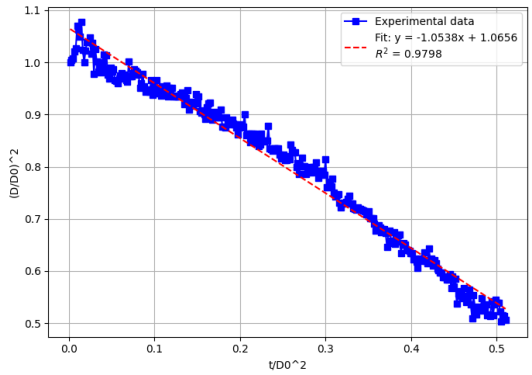
tinuous, overcoming the disruptions seen at lower concentrations. At the highest loading of 30% Al (Fig. 5.4d), regression maintains a small burn time.



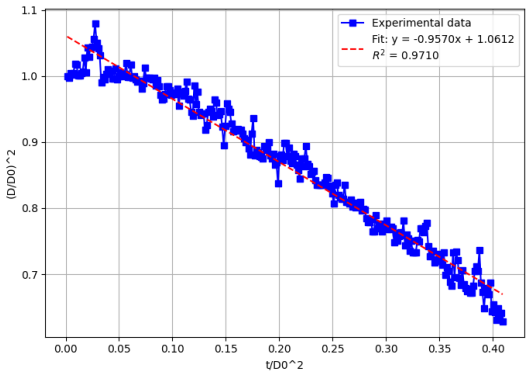
(a) 5% SiO_2, 0% Al



(b) 5% SiO_2, 10% Al



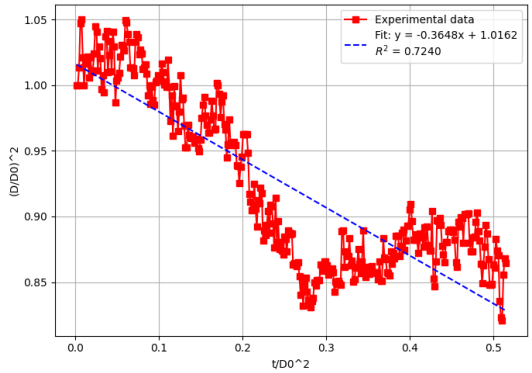
(c) 7.5% SiO_2, 0% Al



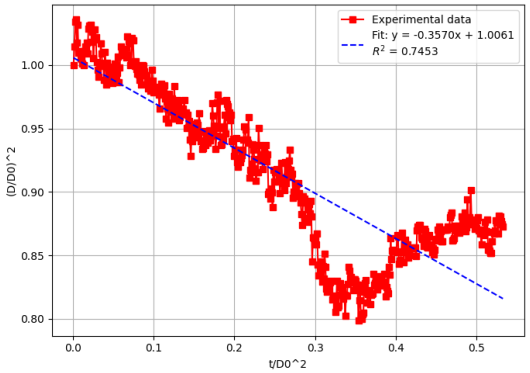
(d) 7.5% SiO_2, 10% Al

Figure 5.3: Effect of silica and aluminum loading on inorganic gel propellant characteristics

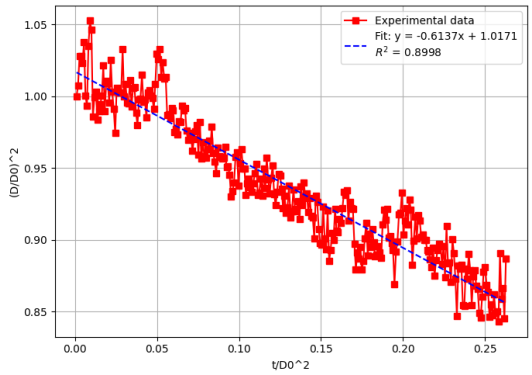
Figure 5.5 presents the D^2 law droplet burning for gel propellants where the mass ratio of the gellant (Thixotrol ST) and solvent (Xylene) are in equal ratios, ranging from 5% to 7.5%. Similarly, a 15% aluminum content is also used with the highest concentration. Across all non-aluminized cases, the droplet combustion shows linearity, with (R^2) value above 0.98 which shows a highly steady state burning process. Also, increasing the Thixotrol/Xylene content appears to small increase the burning rate. The effect of metal addition is studied by comparing the pure 7.5% base gel (Fig. 5.5c) with its aluminized part containing 15% Al (Fig. 5.5d). On adding aluminum, the burning rate is increased.



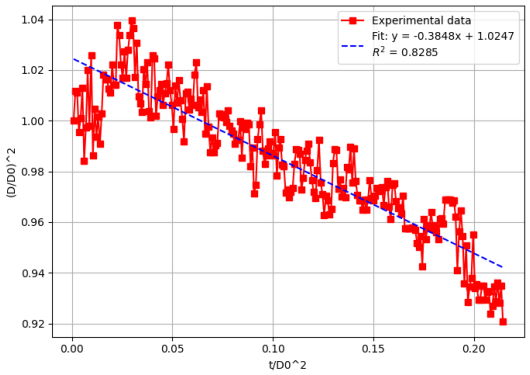
(a) 10% SiO_2, 10% Al



(b) 10% SiO_2, 15% Al

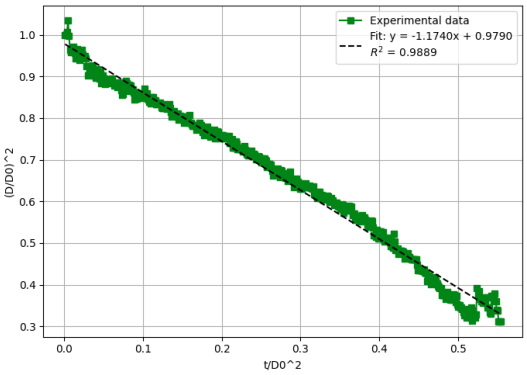


(c) 10% SiO_2, 25% Al

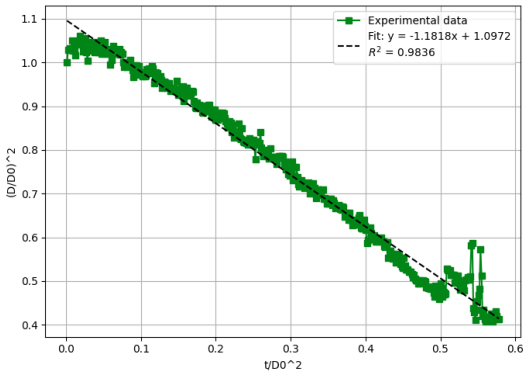


(d) 10% SiO_2, 30% Al

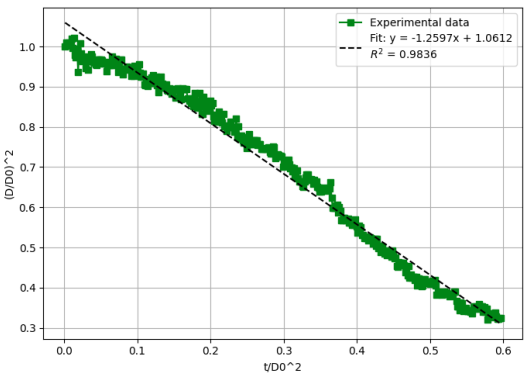
Figure 5.4: Effect of aluminum mass fraction on inorganic gel propellant at fixed silica concentration (10% SiO_2)



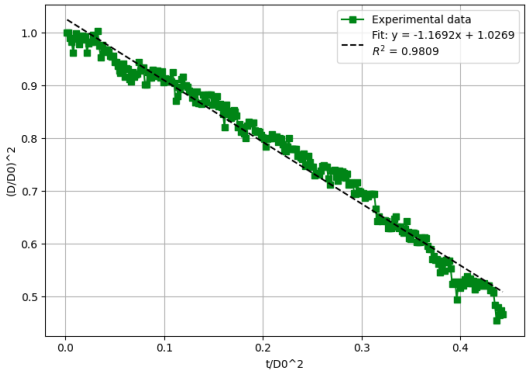
(a) Thixotrol ST 5%, Xylene 5%, Al 0%



(b) Thixotrol ST 6.5%, Xylene 6.5%, Al 0%



(c) Thixotrol ST 7.5%, Xylene 7.5%, Al 0%



(d) Thixotrol ST 7.5%, Xylene 7.5%, Al 15%

Figure 5.5: Effect of aluminum fraction with equal Thixotrol ST and Xylene percentages

6. Hybrid Motor Testing

6.1 Background

A hybrid rocket is a propulsion system where the fuel and oxidizer are stored in different physical states, typically a solid fuel and a liquid oxidizer. This separation makes hybrid rockets relatively inert until the oxidizer is introduced, which is usually done remotely, thereby offering a significant safety advantage compared to conventional solid rockets. With the addition of a proper control valve, hybrid rockets can be throttled or even shut down during operation, similar to liquid rockets, but with the added simplicity of managing only one liquid component, which reduces plumbing complexity. Among the various oxidizers available, nitrous oxide is commonly used in hybrid rocket systems due to its self-pressurizing capability, which eliminates the need for complex pressurization systems and enhances reliability. Although the hybrid engine offer the several advantages it has typically lower combustion efficiency[4]. Typically the combustion efficiency is expressed in the form the characteristics velocity and specific impulse. All self-pressurizing nitrous oxide hybrid engine will have similar thrust profiles, where the thrust is dominated by the oxidizer flow. Unlike a solid rocket motor, varying the geometry of a propellant grain will have very little effect on the engine thrust curve, only the efficiency of the burn.



Figure 6.1: N_2O Regulator and Tank



Figure 6.2: Cured grain (HTPB + TDI)



Figure 6.3: HTPB-N2O Hybrid Engine Test

6.2 Description

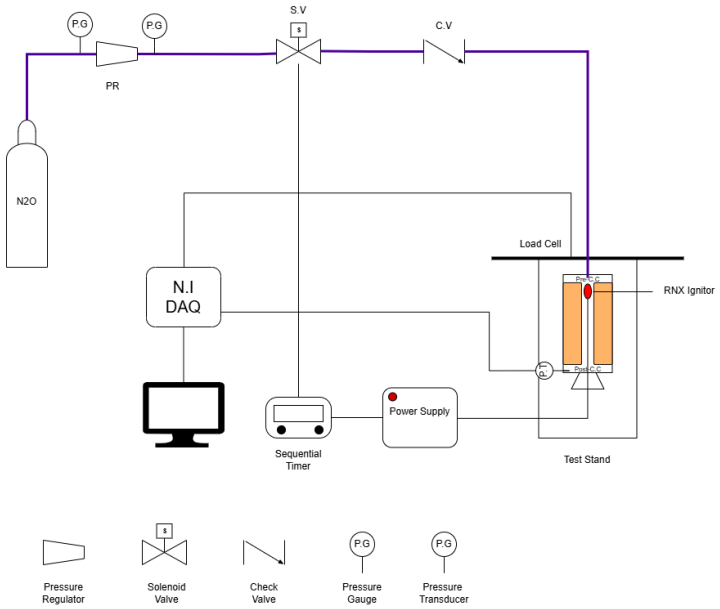
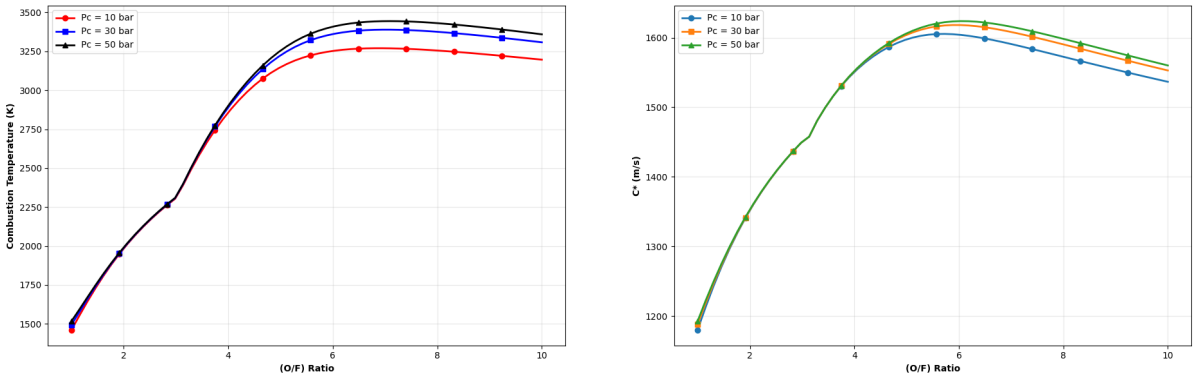


Figure 6.4: Schematic View of Hybrid Engine Experimental Setup

The propulsion system studied in this work is a hybrid rocket engine using hydroxyl-terminated polybutadiene (HTPB) as the solid fuel and nitrous oxide (N_2O) as the oxidizer. In a hybrid engine, oxidizer is injected into a combustion chamber containing a solid fuel grain with a core where combustion and regression of the fuel occurs. After the ignition nitrous oxide flows through the injector into the port, decomposes and reacts with vaporized HTPB, generating high-temperature combustion gases that expand through a converging–diverging nozzle to produce thrust. The regression of the given fuel surface is governed by oxidizer mass flux, making injector design and oxidizer distribution important parameters for engine performance. The engine sytem consisted of an oxidizer tank and feed system, modular injector head, fuel grain and chamber casing, ignition system, and nozzle, all mounted on a static test bench equipped with load-cell-based thrust measurement and calibrated pressure transducers connected to a data acquisition system. As part of the work, involvement was made in supporting and observing the assembly and testing of the engine on the test bench, inclcuding, chamber and nozzle assembly, feed line connection, sensor mounting. Two injector configurations were tested and compared: a showerhead injector and a swirl injector. The showerhead injector employed multiple small straight holes distributed across the injector face to produce several parallel oxidizer jets, providing relatively uniform but low-rotation flow into the fuel port. In other hand, the swirl injector introduced tangential velocity components to the oxidizer flow, generating a rotating flow field that increase atomization, increases mixing quality, which can lead to higher regression rates and improved combustion efficiency. Both in-

jector types were tested under similar geometric and operating conditions. In addition to experimental testing, thermochemical analysis of the HTPB–N_2O propellant combination was performed using chemical equilibrium tools to estimate theoretical performance parameters, including adiabatic flame temperature, characteristic velocity (c^*), specific impulse (I_{sp}), over a range of o/f ratio. For the ignition system typical pyrotechnic igniter was used which is made up of RNX propellant(Potassium Nitrate, Epoxy Resin and Red Iron Oxide in required ratio). The typical system for the hybrid test consists of assembly of the multiple components as shown in Fig 6.4. The ignition sequence was controlled with the timer system managed with sequential timer.

The ignition was done electrically with the help of DC power supply system. When the ignition switch is pressed the igniter get energized. When the igniter is burned after the certain delay the solenoid valve is opened which feed the nitrous oxide at a required feed pressure to the injector and after the combustion of the engine occurs then the solenoid is closed automatically by the sequential timer system at a required delay time. The feed pressure for the nitrous oxide is set based on the pressure drop consideration along the feed line and the different kind of control valve. A check valve is also installed before the injector inlet. Fig 6.5 shows the typical performance parameter obtained from the chemical equilibrium analysis tools.



(a) Combustion temperature vs O/F

(b) C* vs O/F

Figure 6.5: Thermal and performance parameters with mixture ratio for HTPB–N_2O propellant.

7. Sieving of Ammonium Perchlorate (NH_4ClO_4)

Sieving of ammonium perchlorate (AP) is the process of separating AP particles into different size ranges using standard mesh sieves. This is mainly done to obtain a uniform particle size distribution, which is essential in solid propellant manufacturing because particle size significantly affects the burning rate, packing density, and overall combustion performance of the propellant. Uniform AP particles also improve the quality of mixing with binder and other additives, resulting in a more consistent propellant grain.

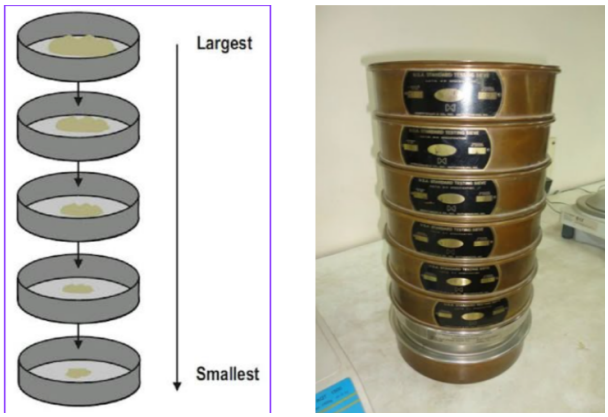


Figure 7.1: Set of sievs for different particle sizes

In this process, AP powder is placed on a set of sieves arranged from larger to smaller mesh sizes and then shaken using tap-sieve shaker machine for a fixed time. Larger particles remain on the upper sieves, while smaller particles pass through to the lower sieves, thus separating the AP into coarse, medium, and fine fractions. After sieving, each size fraction is collected, labeled, and stored separately. Since AP is a strong oxidizer, proper safety measures such as wearing gloves and masks, avoiding contamination, and minimizing dust formation must be followed.

8. X-Ray Imaging of Solid Propellant

X-ray imaging of solid propellant is a non-destructive testing (NDT) technique used to examine the internal structure of a propellant grain without cutting or damaging it. This method helps in detecting internal defects such as cracks, voids, air pockets, inclusions, debonding, or non-uniform distribution of oxidizer particles. During the process, the solid propellant sample is placed between an X-ray source and a detector or film, and X-rays are passed through it. Since different materials absorb X-rays differently depending on their density, the internal features appear as variations in contrast on the X-ray image. This technique is important for ensuring the safety and reliability of solid rocket motors, as internal defects can lead to abnormal burning and possible failure.

sorbital(35%) + Potassium Nitrate (KNO_3) (65%)

sorbital (a sugar-based fuel) mixed with potassium nitrate (KNO) forms a type of sugar-based solid propellant system where potassium nitrate acts as the oxidizer and disorbital acts as the fuel. When ignited, KNO provides oxygen to support the combustion of disorbital, producing hot gases that generate thrust. This combination is commonly studied in basic propellant experiments because it is relatively easy to prepare and demonstrates the fundamental principle of solid propellant combustion. The performance of this propellant depends on factors such as the fuel-to-oxidizer ratio, particle size of KNO, and uniformity of mixing.



Figure 8.1: Preparation of Propellant Grain

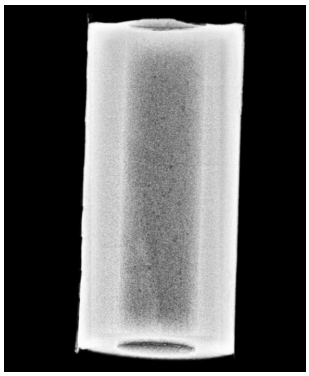


Figure 8.2: X-Ray Image of Propellant Grain

The X-ray image of propellant grain shows a mostly continuous and uniform internal structure, indicating reasonably good mixing and casting. The central region appears slightly darker and grainy, which suggests a small variation in density due to particle packing effects, minor porosity, or non-uniform oxidizer distribution. The darker circular regions at the top and bottom ends are technically significant, as they indicate end voids or shrinkage cavities formed during cooling/solidification, or trapped air due to insufficient degassing. These voids can affect combustion by creating localized hot spots, increasing the burning surface area unexpectedly, and potentially causing unstable burning or pressure rise. Overall, the X-ray confirms that the grain is largely homogeneous but contains minor end defects and density gradients typical in hand-cast sugar-based propellant sample.

9. Conclusion

This internship provided practical experience in experimental propellant combustion and rocket propulsion testing across multiple propellant classes, including composite solid propellants, slurry propellants, gel propellants, electrically controlled solid propellants (ECSP), and hybrid propulsion systems. The work involved propellant preparation, laser ignition experiments, Crawford burner testing, droplet combustion studies, and hybrid motor testing supported by high-quality instrumentation and diagnostic techniques.

The laser ignition experiments demonstrated that ignition delay is strongly dependent on the input laser power, where higher power levels resulted in faster and more reliable ignition. Crawford burner tests conducted on ECSP and slurry propellants were used to characterize burning rate behavior and to study the influence of additives and formulation changes on pressure-dependent regression characteristics.

Gel propellant synthesis and droplet combustion testing confirmed that both organic and inorganic gel systems can achieve stable combustion. The gellant concentration was observed to significantly affect the rheological properties and structural stability of the gel, while aluminum addition increased energy release and enhanced combustion intensity.

Hybrid motor testing provided practical understanding of injector design effects, ignition sequencing, and experimental methods used for thrust and performance measurement.

In the sieving process of ammonium perchlorate (AP), the oxidizer was classified into coarse, medium, and fine particle size fractions using a tap-sieve shaker with sieves arranged in descending mesh sizes. Since AP is a strong oxidizer, strict safety precautions were followed to prevent contamination, minimize dust formation, and ensure safe handling procedures.

X-ray radiography of the propellant grain indicated a mostly uniform internal structure, confirming satisfactory mixing and casting quality. Minor density variations in the central region suggest small porosity or localized non-uniform particle distribution. Darker regions near the ends indicate possible voids or shrinkage cavities caused by trapped air or incomplete degassing, which can affect combustion stability by increasing local burning surface area and producing pressure fluctuations.

Overall, the internship provided strong technical understanding of propellant formulation, combustion phenomena, and standardized experimental testing procedures by integrating theoretical concepts with practical laboratory and propulsion system testing experience.

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